MILITARY SPECIFICATION

CIRCUIT CARD ASSEMBLY: 11732344 SWITCHING REGULATOR

This specification is approved for use by the US Army Research and Development Coxmand (ARRADCOM) and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements and quality assurance provisions for the Circuit Card Assembly: 11732344 Switching Regulator which is the AZ component of the M21 Ballistic Computer Unit, 11732400.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-I-45607	Inspection Equipment, Acquisition, Maintenance and Disposition of
MIL-STD-45662	Calibration System Requirements'

FSC 1220

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-810	Environmental Test Methods

DRAWINGS

ARRADCOM

11732344	Circuit Card Assembly: 11732344 Switching Regulator

Inspection Equipment

11750202	Automatic Card and Unit, Test Set
11750239	Adapter Computer Board - A2
11821457	Fixture Shock, Vertical and Horizontal Plane
11821458	Shock Adapter, Plate Vib Horz-Vert PlanelShock-Z
11821500	Vibration/Shock, Test Fixture - Computer
11821520	Automatic Test Assembly, Central Processor Unit
11821854	Test Fixture Auto Card, Test Set, Computer FAT

Packaging Data Sheet

PI1732344	Packaging of Circuit Card Assembly: 11732344

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 Fabrication. The switching regulator circuit card assembly, herein referred to as the assembly, shall be manufactured in accordance with Drawing 11732344 and drawings pertaining thereto and, when assembled, shall meet the requirements of this specification. (See 4.5.3.1)

3.1.1 Signals. The assembly shall provide the following signals:

	Signal Designation	Signal Nomenclature
a.	+13V	Preregulator output
b.	115V, 400 Hz	Reticle projector motor reference
c.	RP PWR FAIL	Reference fail flag

3.1.2 General specifications. The following provisions of MIL-F-13926 apply: (See 4.5.1)

a. Order of precedence
b. Dimensions and tolerances
c. Inorganic protective surface finishes
d. Part identification and marking
e. Workmananship

3.1.3 Ambient conditions. Standard ambient conditions shall be as follows:

a.	Temperature	73 ± 18°F
b.	Relative humidity	50 percent ± 30 percent
c.	Atmospheric pressure	28.5 + 2.0 - 3.0 in. Hg.

3.2 First article. When specified (see 6.2), the contractor shall furnish sample units for first article inspection and approval (see 4.4 and 6.2).

3.3 Performance. Unless otherwise specified, the assembly shall meet the performance requirements specified herein under standard ambient conditions of 3.1.3.

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3.3.1 +13V output. With the filter (item 1.5 of table I) connected, loads connected as specified, and power applied as specified, the filter output voltage shall be as follows: (see 4.6.2.1)

3.3.1.1 Nominal output voltage. With the load (item 1.1 of table I) connected and the power source (item 2.2 of table I) set to 24.0 ± 1.2 volts direct current (vdc) the filter output voltage shall be 12.0 ± 0.1 vdc and shall have a ripple content not greater than 0.354 volts root mean square (vrms).

3.3.1.2 Line regulation. With the load (item 1.1 of table I) connected and the power source (item 2.2 of table 1) varied from 16.75 vdc to 30.00 vdc, the filter output voltage shall be 12.00 ± 0.36 vdc.

3.3.1.3 Load regulation. With the load (item 1.2 of table I) connected and the power source (Item 2.2 of table I) set to 24.0 ± 1.2 vdc, the filter output voltage shall be 12.0 ± 0.2 vdc and shall have a ripple content not greater than 0.354 vrms.

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TABLE I. Loads, power and signal sources.

Item	Conditions	Characteristics	Connections
1.0	Loads		Connected between the following pins:
1.1	Resistor	2.5 ohms ±5% 100 watts (w)	P1-22A, P1-22B and P1-23A, P1-23B
1.2	Resistor	1.6 ohms ±5% 200 w	P1-22A, P1-22B and P1-23A, P1-23B
1.3	Resistor	1000 ohms ±5% 1/4 w	P1-11A, P1-11B and P1-3A, P1-3B
1.4	Resistor	1000 ohms ±5% 1/4 w	P1-2A, P1-2B and P1-13A, P1-13B
1.5	Filter (low pass)	R = 71.5 ohms ± 10% C = 2.2 microfarad ± 10%	P1-22A, P1-22B and P1-23A, P1-23B
2.0	Power sources		Applied between the following pins with positive terminal to (+):
2.1	22.6 vdc (24 V LAMP)	Tolerance: ±2.0 vdc Ripple: 500mv peak-to-peak (p-p)max	P1-8A, P1-8B, P1-9A, P1-9B(+) and P1-4A, P1-13A, P1-13B
2.2	24.00 vdc (24 V COMP 1)	Tolerance: As specified, Ripple: 2v p-p max	P1-25A, P1-25B(+) and P1-24A, P1-24B
2.3	15.000 vdc (+15 V)	Tolerance: ±0.150 vdc Ripple: 20mv p-p max	P1-6A, P1-6B(+) and P1-13A, P1-13B
2.4	5.2 vdc (+5 V)	Tolerance: ±0.5 vdc Ripple: 500mv p-p max	P1-14A, P1-14B(+) and P1-13A, P1-13B
2.5	-15.000 vdc (-15 V)	Tolerance: ±0.150 vdc Ripple: 20mv p-p max	P1-21A, P1-21B and P1-13A, P1-13B(+)
3.0	Signal source		When specified, applied between the following pins:
3.1	5.000 vrms, 400 Hz	Tolerances: ±30 mv rms and ± 4 Hz	P1-1A, P1-1B and P1-13A, P1-13B(rtn)

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3.3.2 115V, 400 Hz output. With the load (item 1.3 of table I) connected, power (items 2.3, 2.4 and 2.5 of table I) applied, signal source (item 3.1 of table I) applied, and power (item 2.1 of table I) applied not less than 1 second after all other power sources have attained steady state, the output voltage at P1-11A and P1-11B shall be a 115.0 ± 10.0 vrms sine wave that is in phase ± 10 degrees with the signal source. (See 4.6.2.2)

3.3.3 RP PWR FAIL output. With the load (item 1.2 of table I) connected, and power (items 2.1, 2.3, 2.4, and 2.5 of table I) applied, and the signal source (item 3.1 of table I) applied, the output voltage at P1-2A and P1-2B shall be 0.3 ± 0.3 vdc. With the signal source removed and pins P1-1A, P1-1B connected to pins P1-13A, P1-13B, the output voltage at P1-2A and P1-2B shall be 3.3 + 1.0 vdc. (See 4.6.2.3)

3.4 Environmental.

3.4.1 Shock. The assembly shall be capable of operating as specified herein after exposure under the conditions of 3.1.3 to three half sine wave shock pulses of 40 ±4 gravity unit (g) for a duration of 18 ±3 milliseconds (msec) applied in each direction along three mutually perpendicular axes. In addition, the assembly shall be capable of withstanding three half sine wave shock pulses of 100.0 ±10.0 g for a duration of 1.5 ±0.2 msec applied in each direction along three mutually perpendicular axes. (See 4.6.3.1)

3.4.2 Vibration. The assembly shall be capable of operating as specified herein after exposure under the conditions of 3.1.3 to the vibration profile of figure 1. Duration of exposure shall be not less than 80 minutes in each of three mutually perpendicular axes. (See 4.6.3.2)

3.4.3 Operating temperature. The assembly shall be capable of operating as specified herein over the operating temperature range of -25°F to 140°F. (See 4.6.3.3, 4.6.3.6 and 4.6.3.5)

3.4.4 Storage temperature. The assembly shall be capable of operating as specified herein after exposure to storage temperatures ranging from -65°F to +160°F. (See 4.6.3.3., 4.6.3.4 and 4.6.3.5)

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

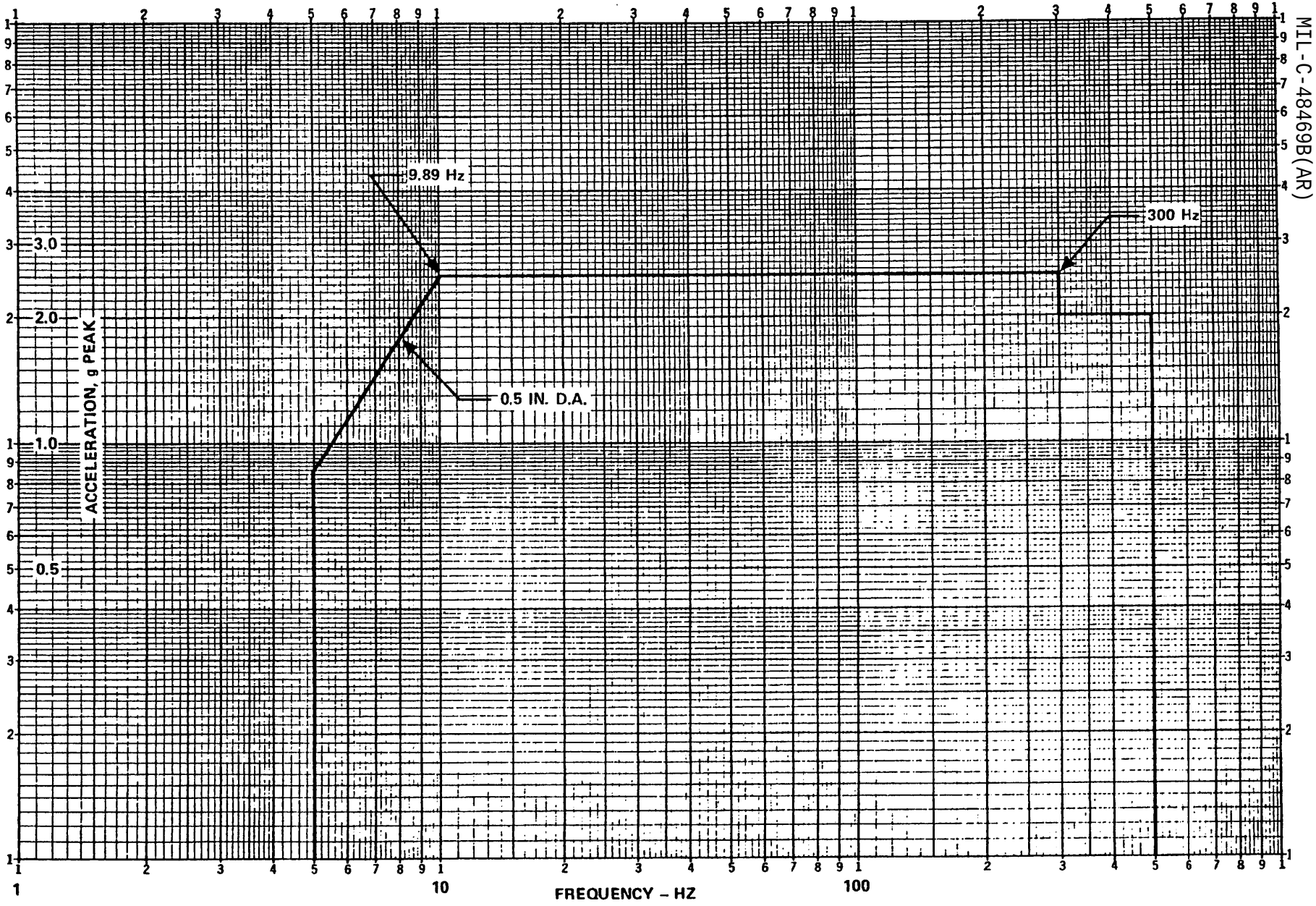
1. First article inspection (See 4.4).
2. Quality conformance inspection (See 4.5).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the quality assurance provisions of MIL-F-13926 and the conditions of 3.1.3.

4.4 First article. The requirement for first article approval and the designation of responsibility for first article inspection to either the Government or the contractor shall be specified in the contract.

4.4.1 Sample. The first article sample shall be three assemblies selected at random by the Government representative from the first fifteen production assemblies.

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4.4.2 Inspection. The sample shall be subjected to all the inspections specified in tables II, III and IV.

4.4.3 Failure. Failure of any assembly to meet any requirement shall be cause for refusal to grant first article approval. The Government reserves the right to terminate first article inspection upon any failure of an assembly to comply with any stated requirement.

4.4.4 Responsibility. The contractor, whether or not responsible, shall inspect the sample for conformance to all contractual requirements and shall submit a record of this inspection with the sample and certificates of conformance for materials. The Government reserves the right to witness inspections performed by the contractor.

TABLE II. Performance tests.

Item	Characteristic	Requirement	Test Procedure
101	+13V output	3.3.1	4.6.2.1
102	115V, 400 Hz output	3.3.2	4.6.2.2
103	RP PWR FAIL output	3.3.3	4.6.2.3

TABLE III. General tests.

Item	Characteristic	Requirement	Test Procedure
301	Fabrication	3.1	Applicable drawings-Visual
302	General Specification	3.1.2	MIL-F-13926 - Visual

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TABLE IV. Environmental tests.

Item	Characteristic	Requirement	Test Procedure
303	Shock	3.4.1	4.6.3.1
304	Vibration	3.4.2	4.6.3.2
305	High Temperature	3.4.3 and 3.4.4	4.6.3.3
306	Low Temperature	3.4.3 and 3.4.4	4.6.3.4
307	Temperature cycling	3.4.3 and 3.4.4	4.6.3.5

4.5 Quality conformance inspection.

4.5.1 Procurement conditions. The following inspection plans shall apply Where the assembly is procured for use as a:

Use	Plan
a. Logistics spare assembly apart from the next higher assembly	A and C
b. Component assembly of the next higher assembly	B and C

4.5.2 Inspection plan A.

4.5.2.1 General sample and tests. One assembly, as a control sample, shall be selected at random by the Government representative from each 100 assemblies produced and shall be subjected to all the tests in table III.

4.5.2.2 Environmental sample and tests. Three assemblies, as a control sample, shall be selected at random by the Government representative from each 50 assemblies produced or from each month's production, whichever occurs first. All the tests in table IV shall be applied separately to each assembly in the sample.

4.5.2.3 Acceptance. Where any assembly of either sample fails to meet any specified requirement, the lot shall be rejected. Rejected lots shall be subject to the provisions of MIL-STD-105.

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4.5.3 Inspection plan B.

4.5.3.1 Sample and tests. The sample shall be selected by the Government representative in accordance with the provisions of MIL-STD-105. All the tests in table III and only the test of item 307 in table IV shall be applied.

4.5.3.2 Acceptance. Acceptance and rejection shall be in accordance with MIL-STD-105.

4.5.4 Inspection plan C.

4.5.4.1 Sample and tests. Each assembly in every lot shall be subjected to all the tests in table II and shall be examined visually for completeness, improper assembly and evidence of poor workmanship.

4.5.4.2 Acceptance. Where any one assembly fails to meet any specified requirement, the detective assembly shall be removed from the lot and resubmitted only after all defects have been corrected.

4.5.5 Inspection equipment. Unless otherwise specified in the contract, the contractor shall supply, maintain and calibrate inspection equipment in accordance with the provisions of MIL-I-45607 and MIL-STD-45662.

4.5.6 Packaging inspection. The sampling and inspection of the preservation - packaging, packing and container marking shall be in accordance with the provisions of packaging data sheet P11732344.

4.6 Methods of inspection.

4.6.1 Test equipment. Unless the assembly is procured apart from the next higher assembly, the test equipment in table V shall be used where specified to perform required tests.

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TABLE V. Test equipment.

Item	Part Number	Nomenclature
1	11750202	Automatic Card and Unit, Test Set
2	11750239	Adapter Computer Board - A2
3	11821457	Fixture Shock, Vertical and Horizontal Plane
4	11821458	Shock Adapter, Plate Vib Horz-Vert Plane/ Shock-Z
5	11821500	Vibration/Shock, Test Fixture - Computer
6	11821520	Automatic Test Assembly, Central Processor Unit
7	11821854	Test Fixture Auto Card, Test Set, Computer FAT

4.6.2 Performance tests.

4.6.2.1 +13V output. Use items 1, 2 and 6 of table V or suitable equipment to verify the requirements of 3.3.1.

4.6.2.2 115V, 400 Hz output. Use items 1, 2 and 6 of table V or suitable equipment to verify the requirements of 3.3.2.

4.6.2.3 RP PWR FAIL output.- Use items 1, 2 and 6 of table V or suitable equipment to verify the requirements of 3.3.3.

4.6.3 Environmental tests.

4.6.3.1 Shock.

4.6.3.1.1 Basic design. Use items 3, 4 and 5 of table V or mount the assembly in a suitable shock fixture and subject the assembly to the shock test specified in MIL-STD-810, method 516.2, procedure I, figure 516.2-2. Apply three half sine wave shock pulses in each direction along the three axes. Peak amplitude shall be 40 ±4 g with a time duration of 18 ±3 msec measured at the 10 percent amplitude points. At the conclusion of this test, subject the assembly to the performance tests of table II.

4.6.3.1.2 High intensity. Use items 3, 4 and 5 of table V or mount the assembly in a suitable shock fixture and subject the assembly to the shock test specified in MIL-STD-810, method 516.2, procedure IV, figure 516.2-2. Apply three half sine wave shock pulses in each direction along the three axes. Peak amplitude shall be 100.0 +10.0 g with a time duration of 1.5 ±0.2 msec measured at the 10 percent amplitude points. At the conclusion of this test, subject the assembly to the performance tests of table II.

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4.6.3.2 Vibration. Vibration A as specified in 4.6.3.2.1 shall be conducted only for first article inspection. Vibration B as specified in 4.6.3.2.2 shall be conducted only for quality conformance inspection.

4.6.3.2.1 Vibration A. Use items 4 and 5 of table V or mount the assembly in a suitable vibration fixture and subject the assembly to the vibration test specified in MIL-STD-810, method 514.2, procedure VIII, except that the test level shall be the vibration curve shown in figure 1 and duration of exposure shall be not less than 80 minutes per axis. At the conclusion of this test, subject the assembly to the performance tests of table II.

4.6.3.2.2 Vibration B. This test shall be conducted as in 4.6.3.2.1 except that the period of vibration shall be not less than 15 minutes in each axis.

4.6.3.3 High temperature. Use item 7 of table V or suitable equipment and subject the assembly to the high temperature test specified in MIL-STD-810, method 501.1 procedure I. The highest operating temperature shall be 140°F. The performance tests of table II shall be applied.

4.6.3.4 Low temperature. Use item 7 of table V or suitable equipment and subject the assembly to the low temperature test specified in MIL-STD-810, method 502.1, procedure I. The storage temperature shall be -65°F. The lowest operating temperature shall be -25°F. The performance tests of table II shall be applied.

4.6.3.5 Temperature cycling. With table I loads 1.0 thorug 1.5 and power 2.0 through 2.5 applied, subject the assembly to the temperature profile specified in figure 2. At the conclusion of this test, subject the assembly to the performance tests of table II.

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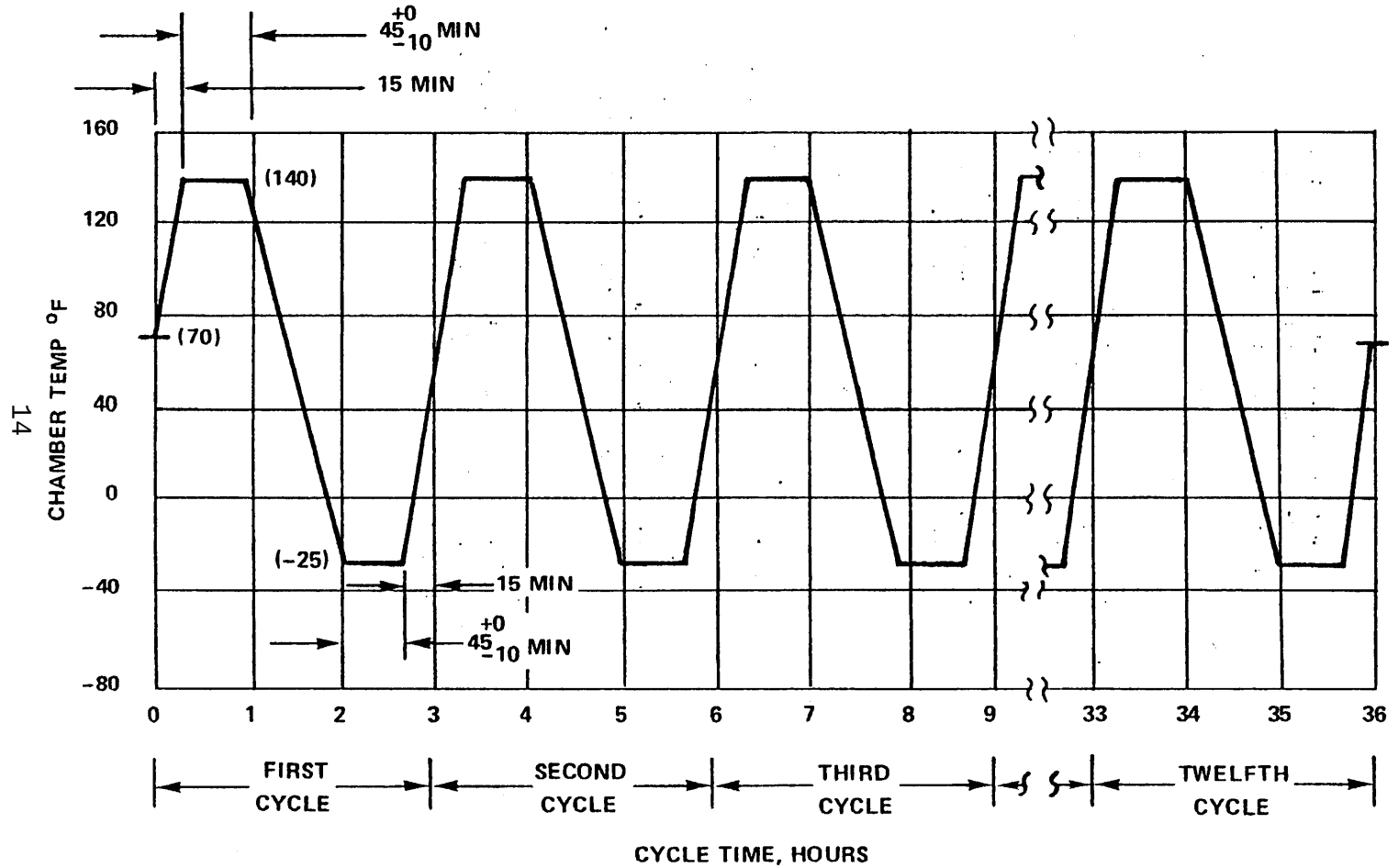


FIGURE 2. Temperature cycling.

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5. PACKAGING

5.1 Packaging, packing and marking. Packaging, packing and marking shall be in accordance with packaging data sheet P11732344. The level of protection shall be as specified in the procurement document. (See 4.5.6)

6. NOTES

6.1 Intended use. The Switching Regulator circuit card assembly is used in the MZ1 Ballistic Computer Unit.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.

b. Special tests and examination as required, the type of inspection required and the activity responsible for testing and its facility.

c. Applicable packaging data sheet number (see 5,1).

d. Selection of applicable levels of preservation, packaging, and packing.

e. Applicable stock number.

f. First article inspection requirements.

6.3 Previous revisions of MIL-C-48469B were for the US Army Armament Research and Development Command's internal use only, and were not forwarded to the Naval Publications and Form Center for publication.

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-C-48469B CIRCUIT CARD ASSEMBLY: 11732344 SWITCHING REGULATOR

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

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